

CPSC 375

Introduction to Data Science and Big Data Analytics

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Week 02 L03

What we will cover today

- Data visualization
- Grammar of graphics
 - Hadley Wickham's layered grammar
 - Lets-Plot in Python
 - Inspired by ggplot2 in R

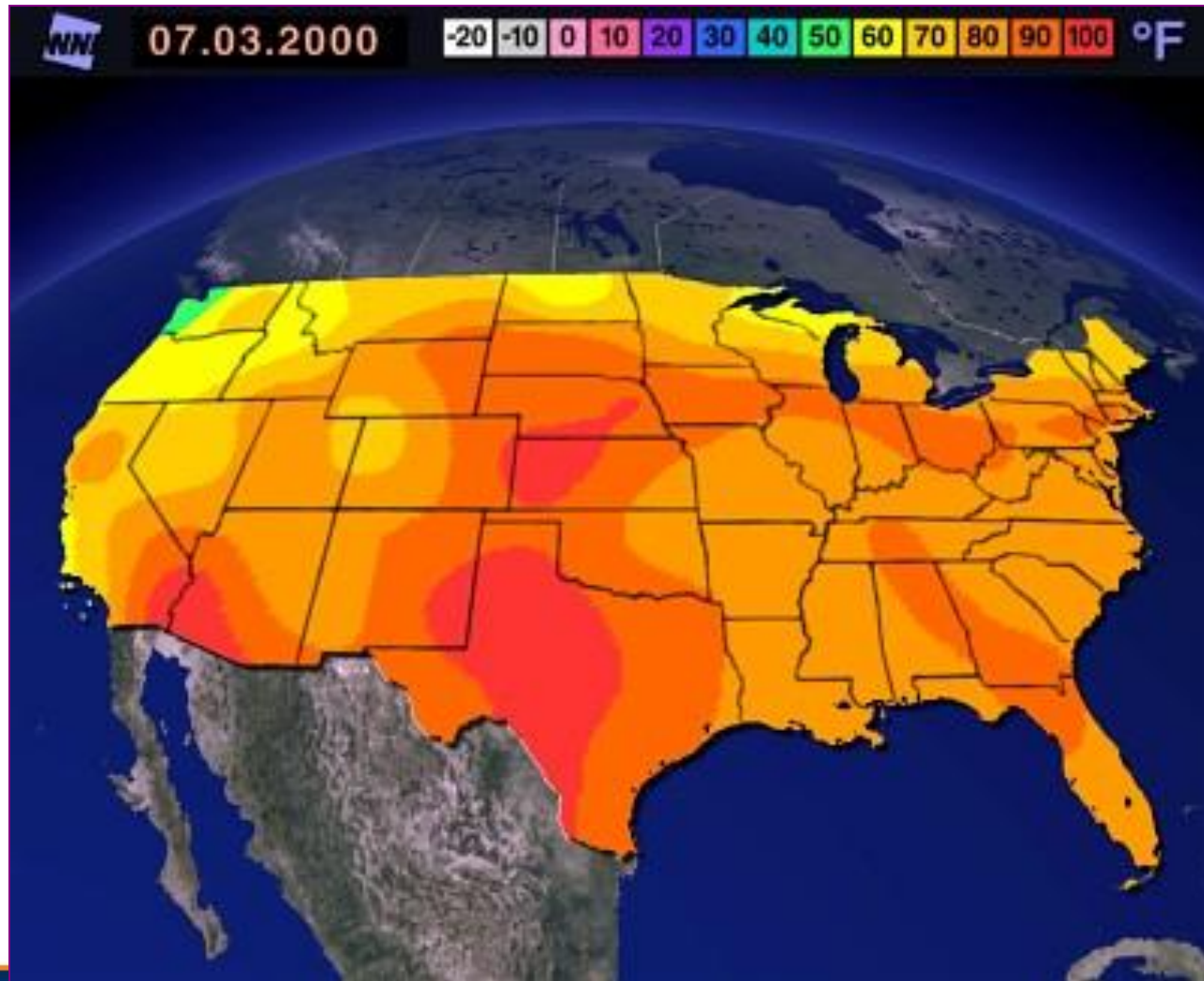
Data Visualization

- Problem
 - Big datasets: How to understand them?
- Solution
 - Take better advantage of human perceptual system
 - Convert information into a graphical representation.
- Issues
 - How to convert abstract information into graphical form?
 - Do visualizations do a better job than other methods?

Goals of Information Visualization

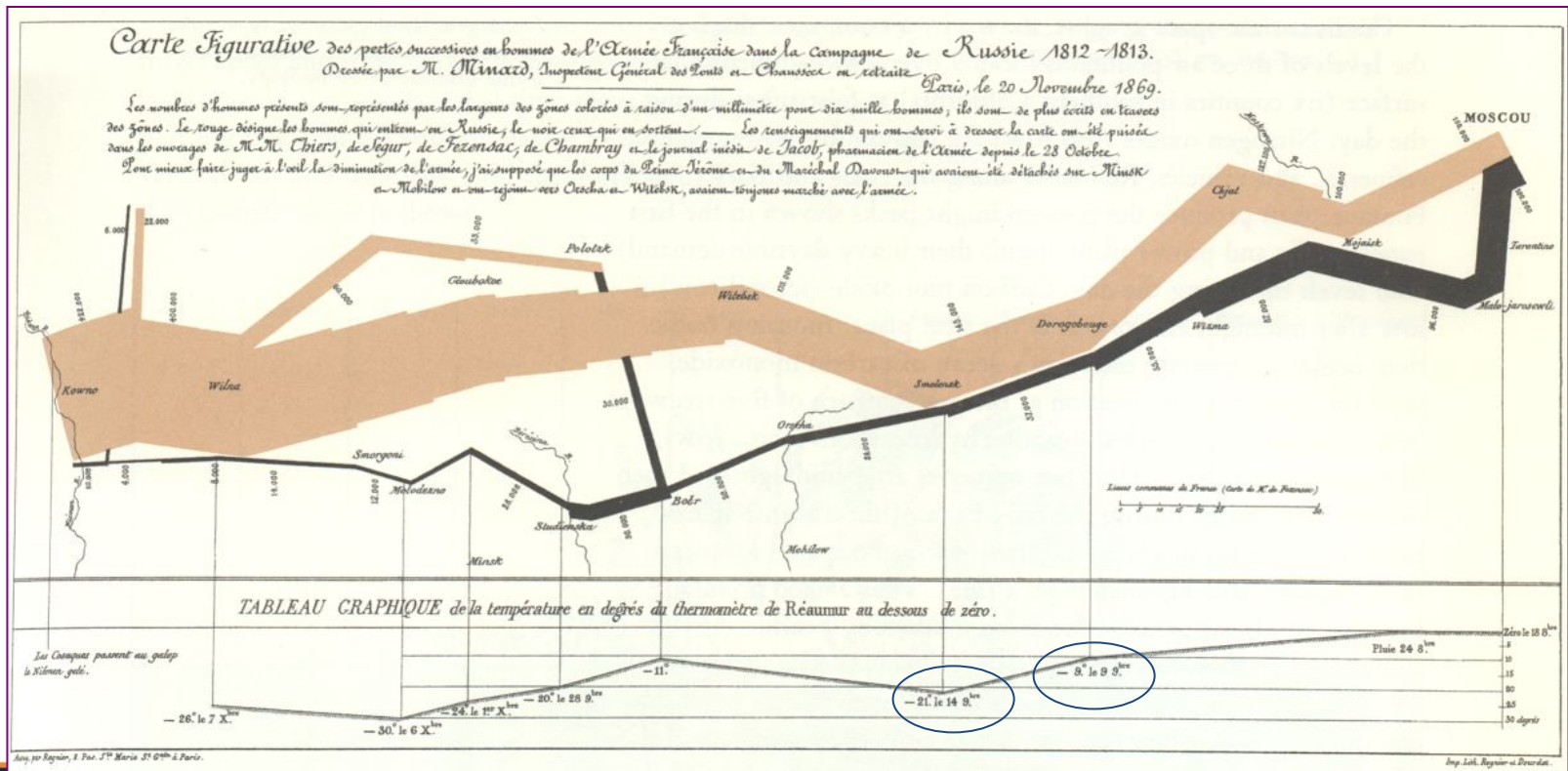
- Present large amounts of data
- Present information from various viewpoints
- Present information at several levels of detail
(from overviews to fine structure)
- Support visual comparisons
- Tell stories about the data

Example: good visualization



Example: good visualization

- The French engineer, Charles Minard (1781-1870), illustrated the disastrous result of Napoleon's failed Russian campaign of 1812.
- Graph shows
 - Size of the army on its outward and return legs by the width of the band across the map of the campaign
 - Temperature on the retreat shown on the line graph at the bottom.

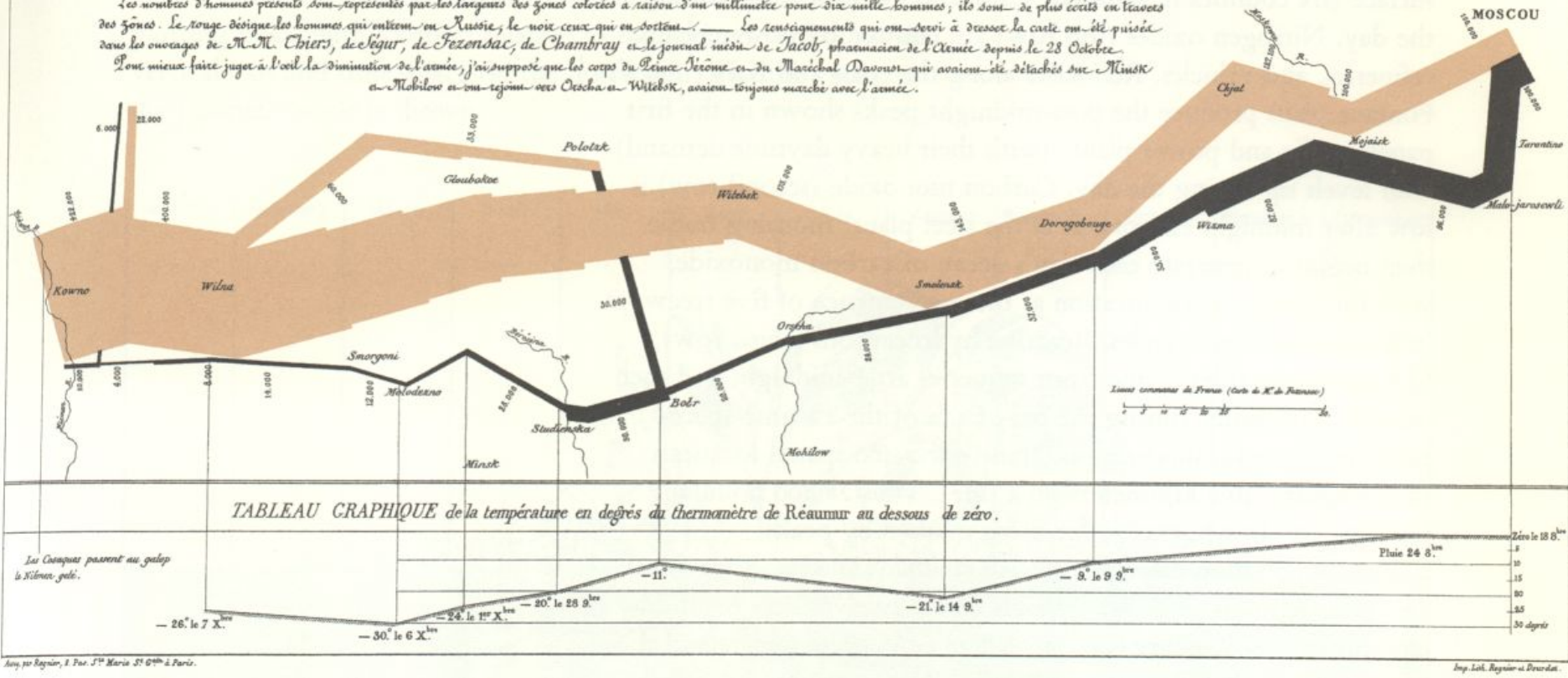


Napoleon's 1812 March by Minard

Carte Figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812-1813.

Dressée par M. MINARD, Inspecteur Général des Ponts et Chaussées en retraite Paris, le 20 Novembre 1869.

Les nombres d'hommes présents sont représentés par les longueurs des zones colorées à raison d'un millimètre pour dix mille hommes; ils sont de plus écrits en traits des zones. Le rouge désigne les hommes qui entrent en Russie; le noir ceux qui en sortent. Les renseignements qui ont servi à dresser la carte ont été puisés dans les ouvrages de M. M. Chiers, de Leguay, de Fezensac, de Chambray et le journal inédit de Jacob, pharmacien de l'Armée depuis le 28 Octobre. Pour mieux faire juger à l'œil la diminution de l'armée, j'ai supposé que les corps du Prince Jérôme et du Maréchal Davoust qui avaient été détachés sur Minsk et Mohilew et qui rejoignent Orescha et Witebsk, avaient toujours marché avec l'armée.



Variables shown:

- size of army
- direction

- latitude
- longitude

- temperature
- date

[Tufte]

Why Visualization?

- Use the eye for pattern recognition; people are good at
 - scanning
 - recognizing
 - remembering images
- Graphical elements facilitate comparisons via
 - length
 - shape
 - orientation
 - texture

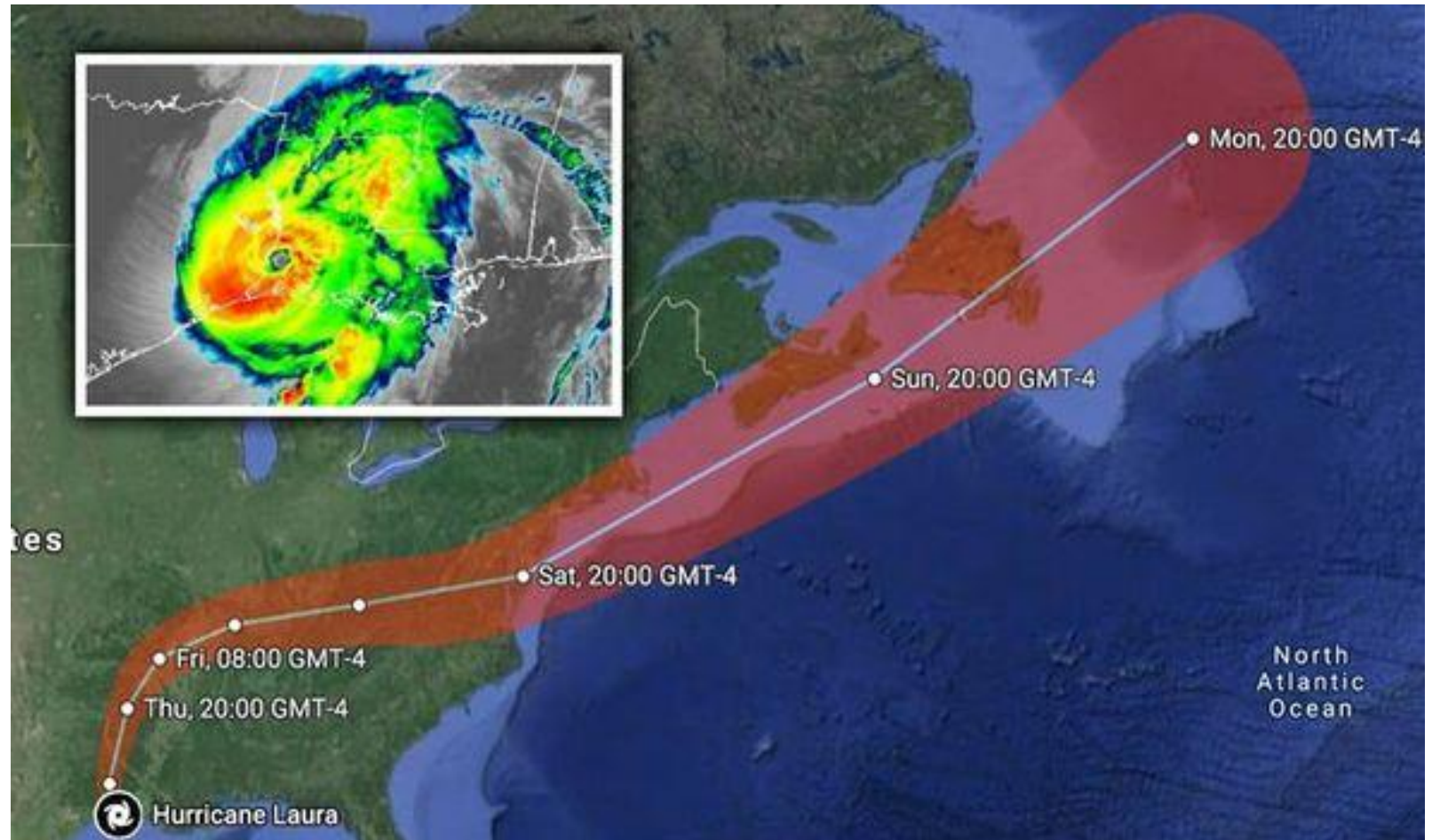
Why visualization?

“The simple graph has brought more information to the data analyst’s mind than any other device.” — John Tukey

Two Different Goals of Visualization

- Communicate
 - Explain
 - Make Decisions
- Explore/Calculate
 - Analyze
 - Reason about Information

Which is better?

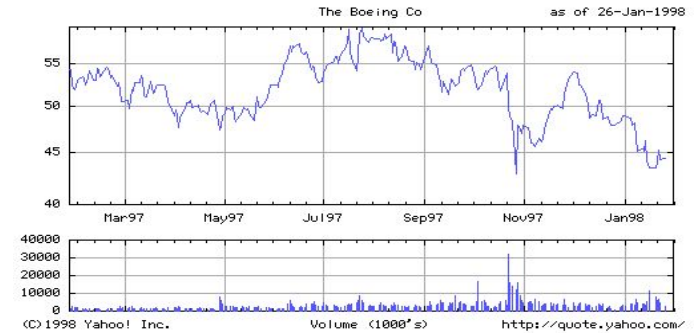


Which is better?

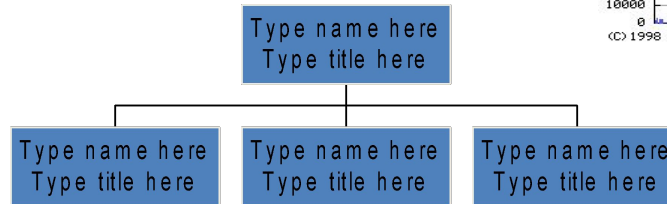


Types of Symbolic Displays

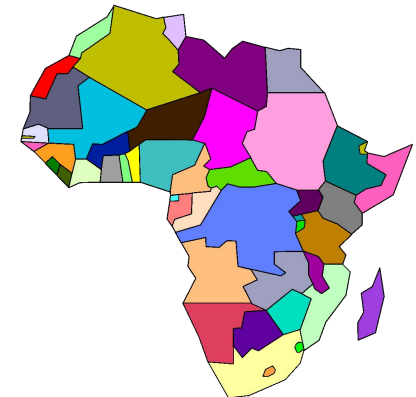
- **Graphs**



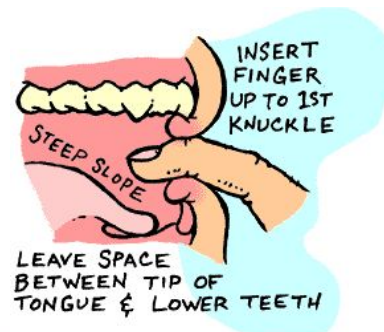
- **Charts**



- **Maps**



- **Diagrams**



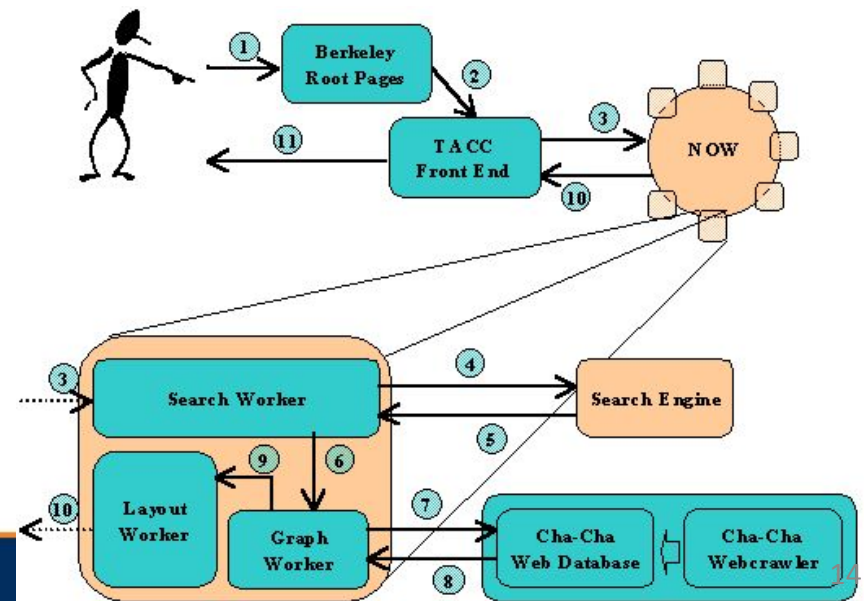
Types of Symbolic Displays

Charts

- discrete relations among discrete entities
- structure relates entities to one another
- lines and relative position serve as links

- **Examples:**

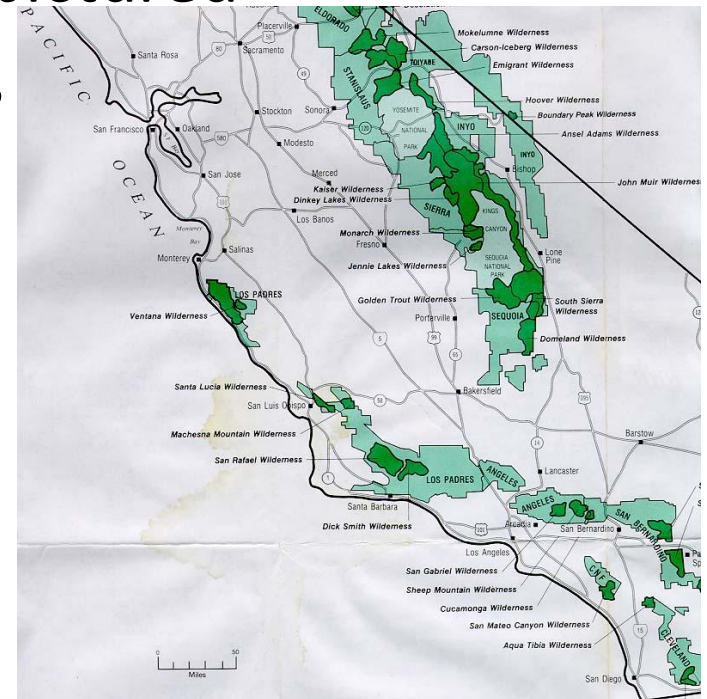
family tree,
flow chart,
network diagram



Types of Symbolic Displays

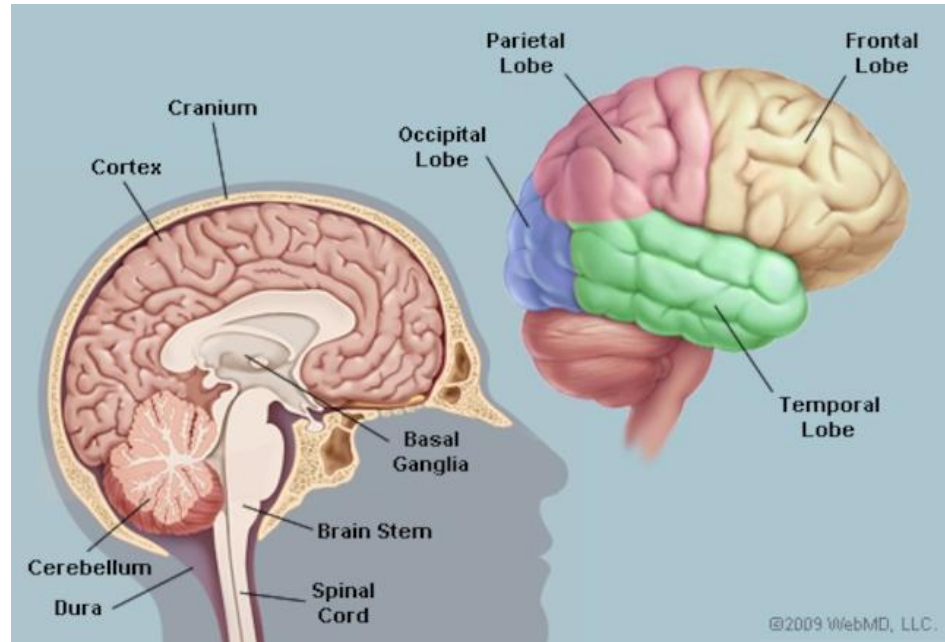
Maps

- internal relations determined (in part) by the spatial relations of what is pictured
- labels paired with locations
- **Examples:**
 - physical maps,
 - topographic maps,
 - political maps,
 - maps of census data



Types of Symbolic Displays

- Diagrams
 - schematic pictures of objects or entities
 - parts are symbolic (unlike photographs)
 - Examples:
 - how-to illustrations,
 - figures in a manual



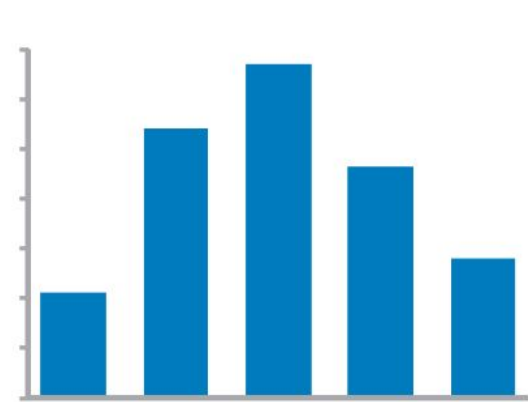
Types of Symbolic Displays

Graphs

- at least two scales required
- values associated by symmetric “paired with” relation
- **Examples:** scatterplot, bar graph



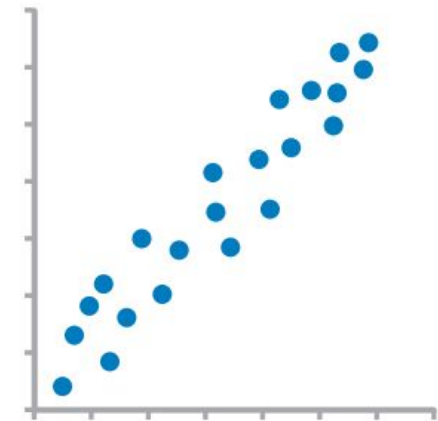
Some Common Graph Types



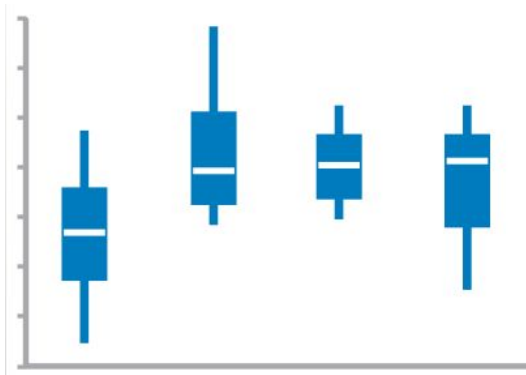
Bar Graph



Line Graph



Scatter Plot



Box Plot

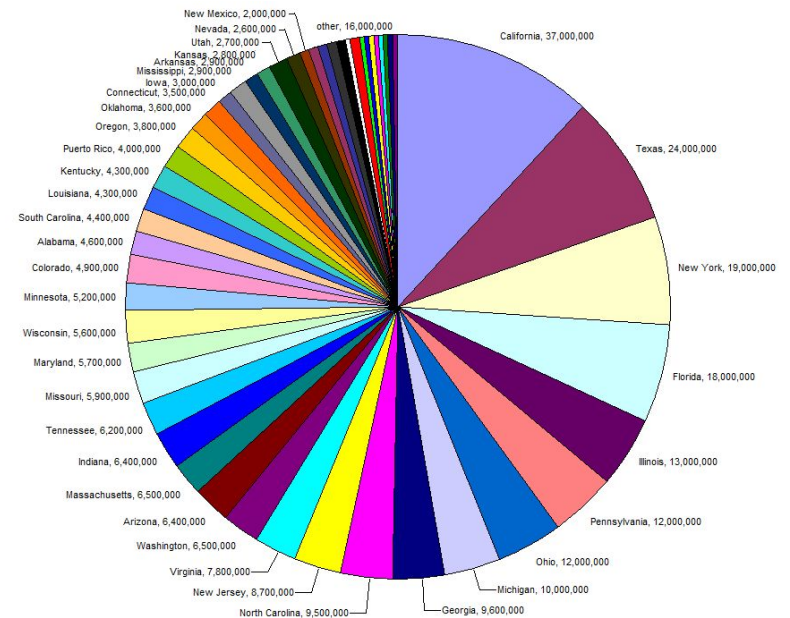
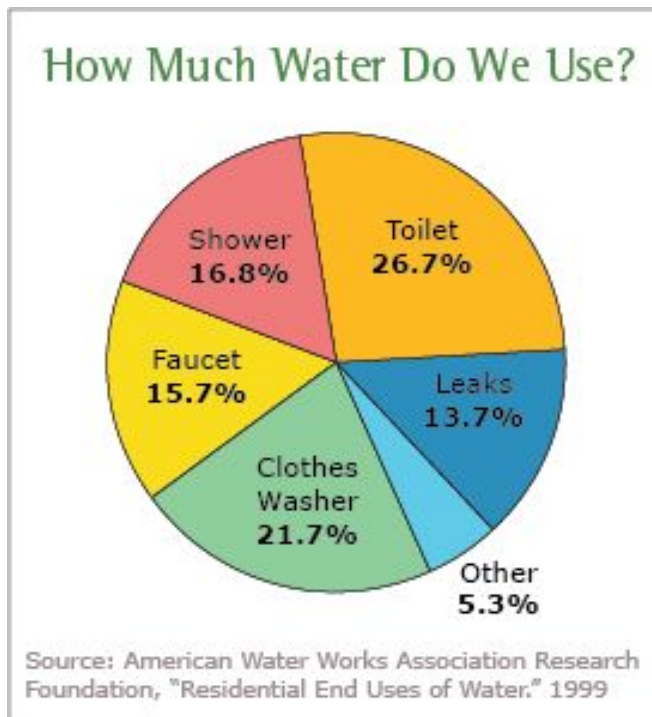
Kim Unger, kunger@databrains.com

When to use which type?

- Line graph
 - Variables must have continuous values
 - conventional ordering among ordinals
- Bar graph
 - compare relative point values
- Scatter plot
 - convey overall relationship between two variables
- Histogram, box plot
 - Show distribution of continuous values

When to use which type?

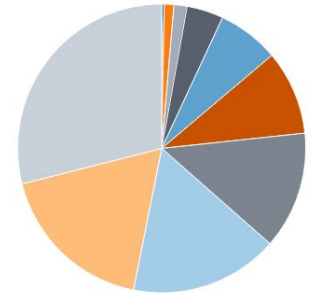
- Pie Chart?



WHAT ABOUT PIE CHARTS?

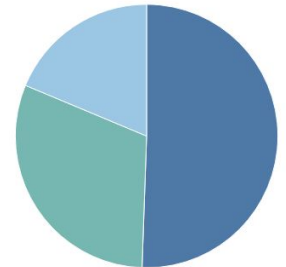
Commonly used to show parts of a whole

Emphasizing differences in proportion among a few numbers



However...

- ❑ Hard to judge relative size of pie slices – **better at differentiating length**
- ❑ Take up a lot of space to **present little information**
- ❑ **Require labels and good color contrast** to even be usable (often difficult)



Best used when one overwhelmingly larger value than the rest – no need to focus on actual values

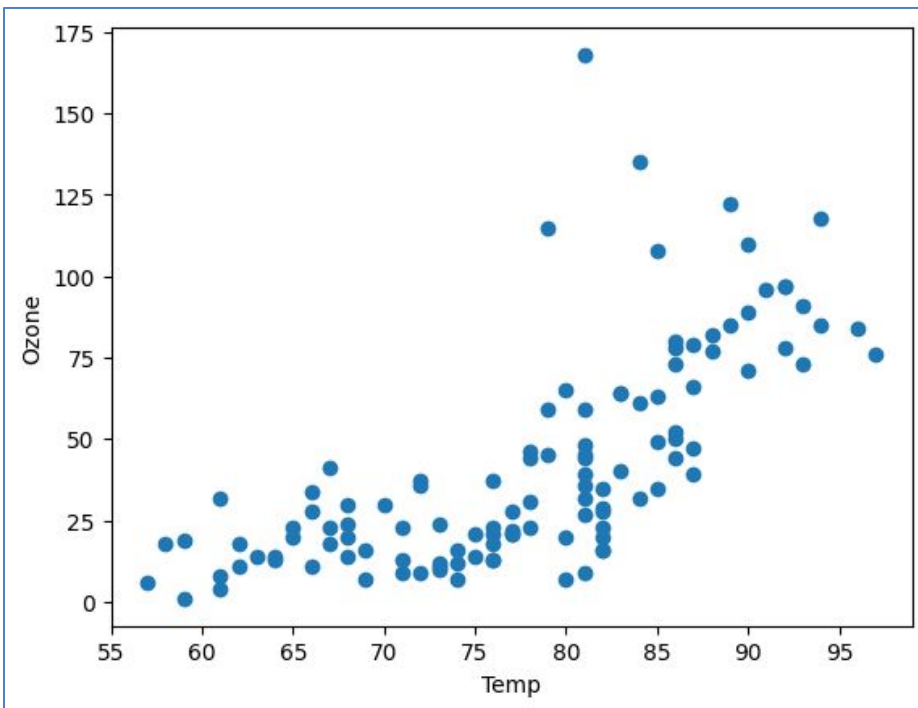
What is common across these types of graphs?

- grammar: “the fundamental principles or rules of an art or science”
- A grammar provides a strong foundation for understanding a **diverse range** of graphics

Python graphics – matplotlib & lets_plot

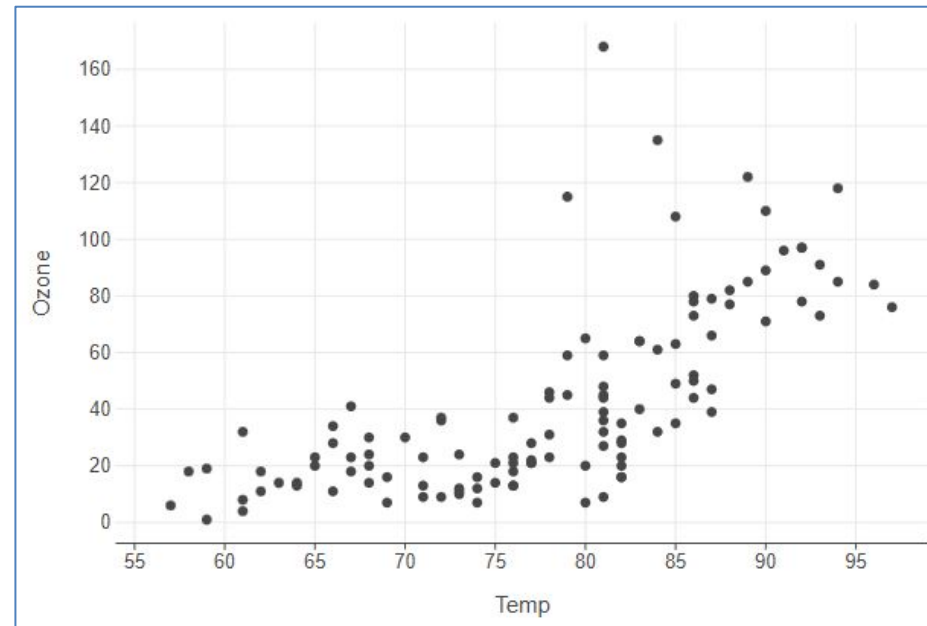
matplotlib:

```
import matplotlib.pyplot as plt
plt.scatter(airquality.Temp, airquality.Ozone)
plt.xlabel('Temp')
plt.ylabel('Ozone')
plt.show()
```

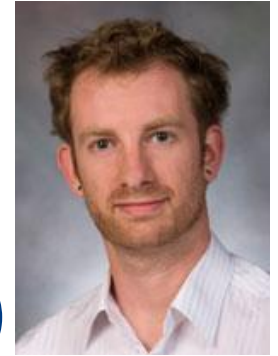


lets_plot:

```
ggplot(airquality, aes('Temp', 'Ozone')) +
  geom_point()
```



ggplot philosophy



Written by Hadley Wickham (Rice Univ.)

Extends *The Grammar of Graphics* (Wilkinson, 2005)

All graphs can be constructed by combining specifications with data (Wilkinson, 2005).

A *specification* is a structured way to describe how to build the graph from *geometric objects* (points, lines, etc.) projected on to *scales* (x, y, color, size, etc.)

ggplot philosophy

When you can describe the content of the graph with the grammar, you don't need to know the name of a particular type of plot...

Scatterplot, bar graph are just special cases of this formal grammar.

...a plotting system with good defaults for a large set of components that can be combined in flexible and creative ways...

Components of the layered grammar of graphics

- Layer
 - Data
 - Mapping
 - Statistical transformation (stat)
 - Geometric object (geom)
 - Position adjustment (position)
- Scale
- Coordinate system (coord)
- Faceting (facet)
- Defaults

Building a plot in **ggplot** / **Lets-Plot**

data to visualize (a data frame)

map variables to **aes**thetic attributes

geometric objects – what you see (points, bars, etc)

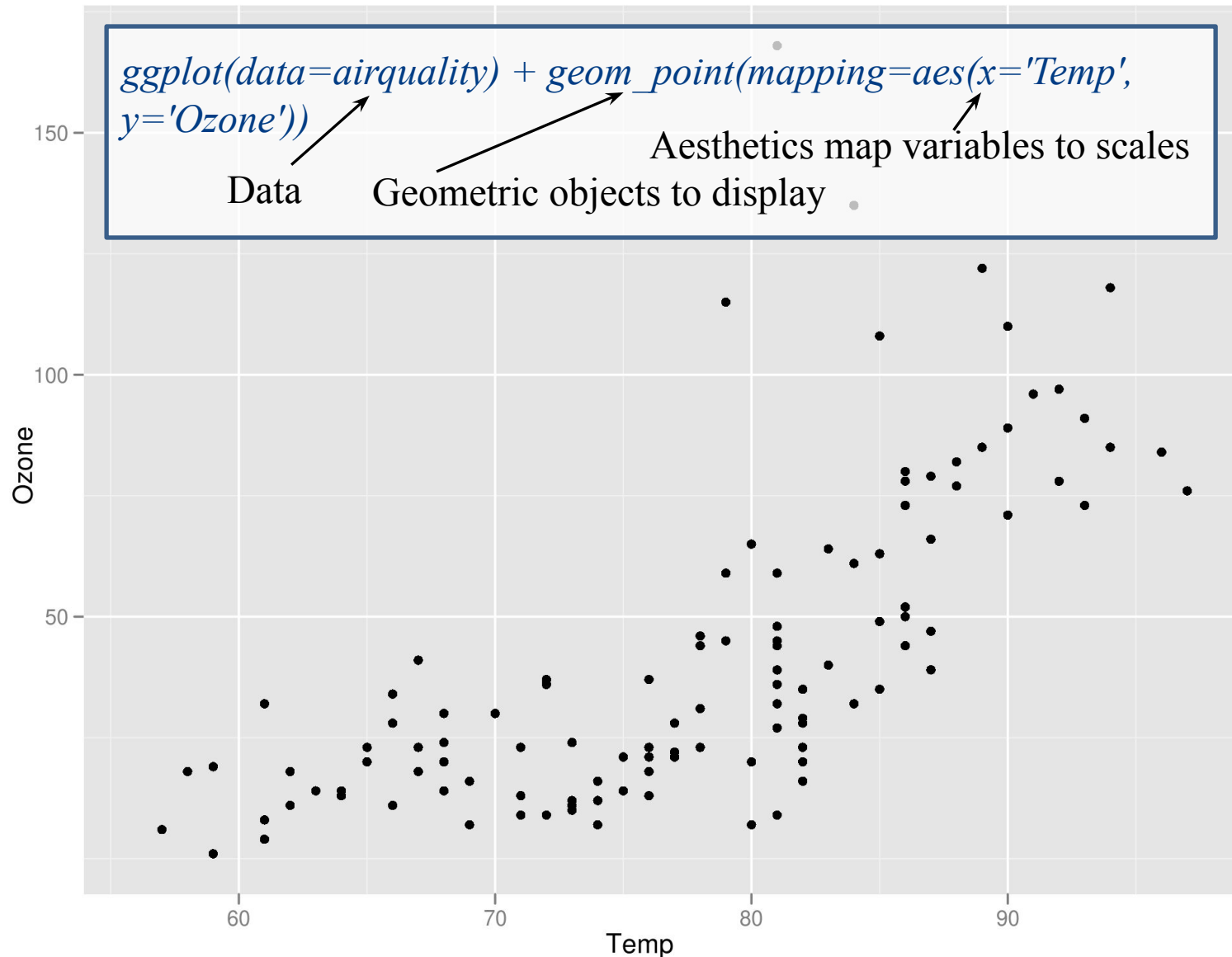
statistical transformations – summarize data

scales map values from data to aesthetic space

faceting subsets the data to show multiple plots

coordinate systems put data on plane of graphic

A basic Lets-Plot graph



Lets-Plot: data

ggplot expects a Pandas **DataFrame**:

Rows: observations

Columns: variables

Different layers can work with different data

Lets-Plot: geometric objects

geoms can be simple (point, line, polygon, bar) or built from these components (boxplot, histogram, ...)

 geom_abline Line, specified by slope and intercept	 geom_area Area plots	 geom_linerange An interval represented by a vertical line	 geom_path Connect observations, in original order
 geom_bar Bars, rectangles with bases on y-axis	 geom_bin2d Add heatmap of 2d bin counts	 geom_point Points, as for a scatterplot	 geom_pointrange An interval represented by a vertical line, with a point in the middle
 geom_blank Blank, draws nothing	 geom_boxplot Box and whiskers plot	 geom_polygon Polygon, a filled path	 geom_quantile Add quantile lines from a quantile regression
 geom_contour Display contours of a 3d surface in 2d	 geom_crossbar Hollow bar with middle indicated by horizontal line	 geom_rect 2d rectangles	 geom_ribbon Ribbons, y range with continuous x values
 geom_density Display a smooth density estimate	 geom_density2d Contours from a 2d density estimate	 geom_rug Marginal rug plots	 geom_smooth Add a smoothed condition mean.
 geom_errorbar Error bars	 geom_errorbarh Horizontal error bars	 geom_text Textual annotations	 geom_segment Single line segments
 geom_freqpoly Frequency polygon	 geom_hex Tile the plane with hexagons	 geom_vline Line, vertical	 geom_step Connect observations by stairs
 geom_histogram Histogram	 geom_hline Line, horizontal	 geom_tile Tile plot as densely as possible, assuming that every tile is the same size.	
 geom_jitter Points, jittered to reduce overplotting	 geom_line Connect observations, in ordered by x value		

Lets-Plot: aesthetics

aesthetics map variables in the data to **visual properties** of geoms

aesthetics include:

x, y position

color, fill,

shape, size, linetype,

alpha, group,

(depending on the geom)

Lets-Plot: aesthetics

Mapping: variable $\leftrightarrow$ visual property

Done within call to `aes(x, y, ...)` (inside the `geom_point`)

```
ggplot(data=airquality) +  
  geom_point(aes(x='Temp', y='Ozone', color='Month'))
```

Color is mapped to month

Can also set visual property to a fixed value (outside the mapping)

```
ggplot(data=airquality) +  
  geom_point(aes(x="Temp", y="Ozone"), color="blue")
```

Color is set to "blue" – not looking for a variable named "blue"

Classwork

- [soodk6/cpsc375-classwork](#)
–visualization.ipynb – Part 1

Mapping discrete and continuous variables

	Discrete	Continuous
Color	“rainbow” of color	Gradient (red to blue)
Size	Increases in set amounts	Smoothly increasing size
Shape	A different shape for each distinct value	Not possible

Lets-Plot: facets

facets are subsets of the data to be displayed next to each other as "small multiples"

- `facet_wrap(facetvar)`
wrap a 1D ribbon of plot panels into a 2D space
can specify `ncol = #`, `nrow = #`
- `facet_grid(rowvar, columnvar)`

scales control whether shared or independent scales
"fixed" (default)

Also possible: "free_x", "free_y", "free"

facet iris

scatterplot by Species

```
(  
  ggplot(data=iris) +  
    geom_point(mapping=aes(x='Sepal.Length', y='Sepal.Width')) +  
    facet_wrap('Species')  
)
```

```
(  
  ggplot(data=iris) +  
    geom_point(mapping=aes(x='Sepal.Length', y='Sepal.Width')) +  
    facet_wrap('Species', scales="free")  
)
```

Lets-Plot: statistical transformations

Stats are transformations that summarize the data

Each stat has a default geom and vice-versa

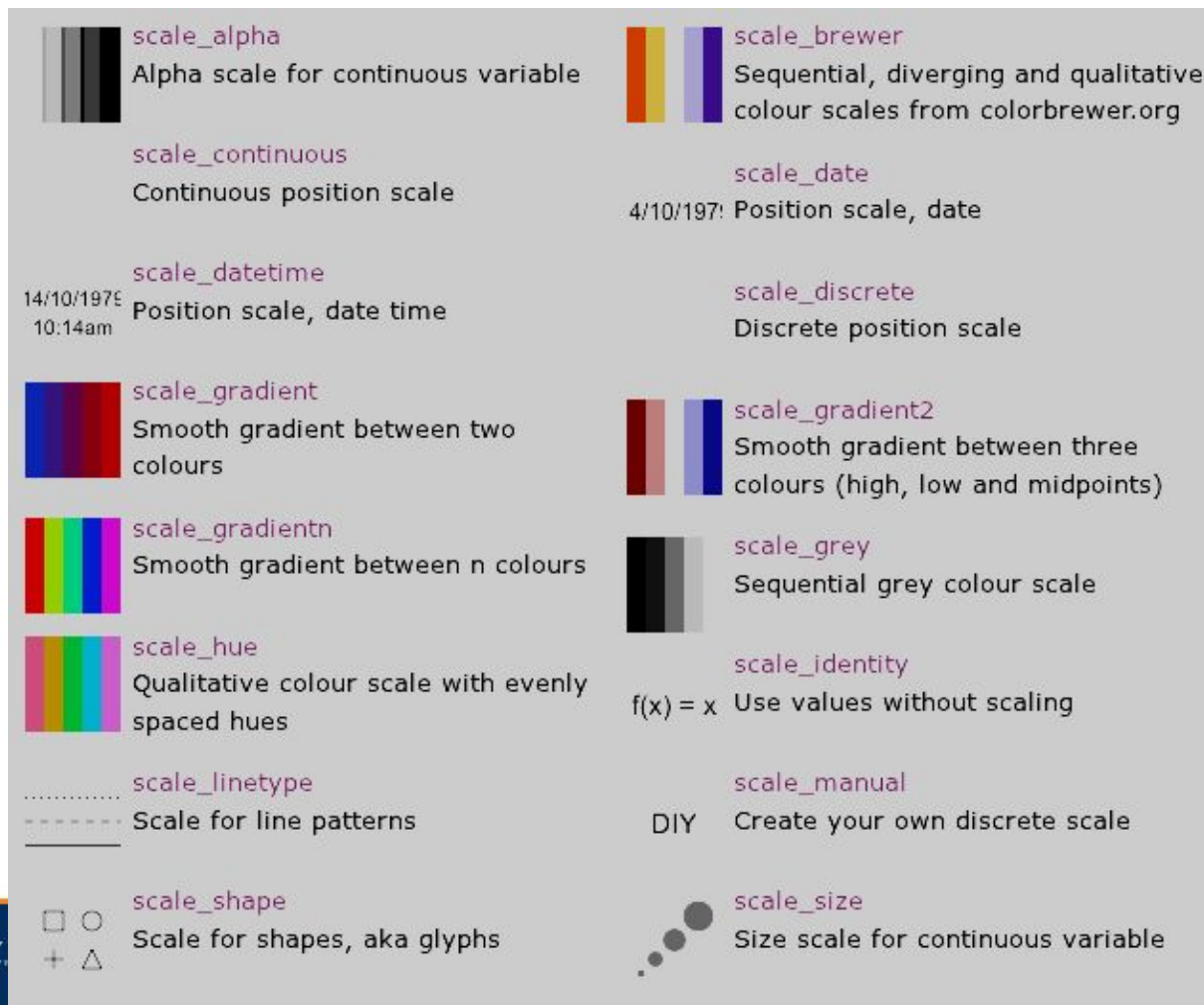
Geom	Stat (default)
geom_point	"identity"
geom_histogram	"bin"
geom_boxplot	"boxplot"

Classwork

- [soodk6/cpsc375-classwork](#)
–visualization.ipynb – Part 2

Lets-Plot: scales

scales control the mapping between data and aesthetics



Lets-Plot: scale

scales control the mapping between data and aesthetics

- assign a unique level of the aesthetic (e.g., a unique color) to each unique value of the variable
- also add a legend that explains which levels correspond to which values

Lets-Plot: scales

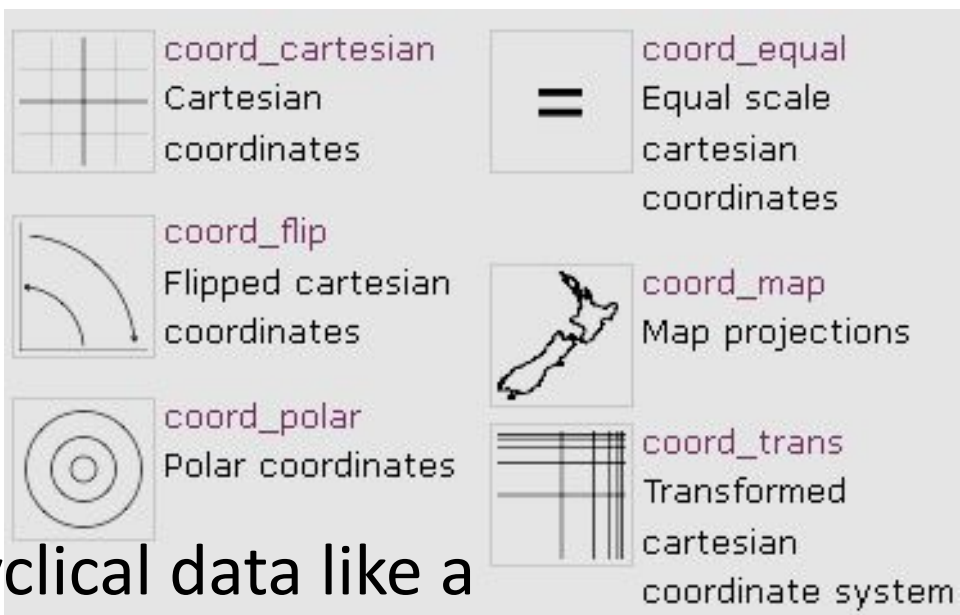
- `ggplot(data=mpg)+geom_point(mapping=aes(x='displ',y='cty')) + scale_x_continuous("Displacement (liters)")`
- `ggplot(data=mpg)+geom_point(mapping=aes(x='displ',y='cty')) + scale_x_continuous("Displacement (liters)", limits=[2,6])`
- `ggplot(data=mpg)+geom_point(mapping=aes(x='displ',y='cty')) + scale_x_log10("Displacement (liters)", limits=[2,6])`
- # shortcut functions
- `ggplot(data=mpg)+geom_point(mapping=aes(x='displ',y='cty')) + labs(x="Displacement", y="Mileage", title="My plot")`
- `ggplot(data=mpg)+geom_point(mapping=aes(x='displ',y='cty')) + labs(x="Displacement", y="Mileage", title="My plot") + xlim(2,6)`

Lets-Plot: coordinate systems

"**coordinate** systems adjust the mapping from coordinates to the 2d plane of the computer screen"

Default is `coord_cartesian()`

Could use `coord_polar()` for cyclical data like a windrose



Lets-Plot: coordinate systems

"**coordinate** systems adjust the mapping from coordinates to the 2d plane of the computer screen"

Default is `coord_cartesian()`

"Other" – a little bit of polish

Themes are sets of specifications for adjustable elements like labels, legends, titles, tickmarks, margins, and backgrounds

```
... + theme_minimal2() # the default look of Lets-Plot  
... + theme_grey() # the default look of ggplot2  
... + theme_bw() # an alternative in black & white
```

Classwork

- [soodk6/cpsc375-classwork](#)
–visualization.ipynb – Part 3

Coming Up

- Data Exploration with plots

Questions?